

# 1000 Genomes Processing README

This README contains information relating to data associated with the 1000 Genomes resequencing done at New York Genome Center.

## Alignment, post-processing and variant calling

Alignment and post-processing are performed exactly as outlined by the Center for Common Disease Genomics project: <https://github.com/CCDG/Pipeline-Standardization/blob/master/PipelineStandard.md>.

## Programs and reference data

The data was aligned to the reference genome using the following programs and reference datasets:

1. [BWA-MEM](#)
2. [Samtools-1.3.1](#)
3. [Picard-2.4.1](#)
4. [GATK-3.5-0](#)
5. Resource files
  - All the resource files used in the analysis can be obtained here: <https://console.cloud.google.com/storage/browser/genomics-public-data/resources/broad/hg38/v0/>.

## Reference genome: GRCh38 with alternative sequences, plus decoys and HLA

The reference genome that the data was aligned to can be obtained here: [ftp://ftp.1000genomes.ebi.ac.uk/vol1/ftp/technical/reference/GRCh38\\_reference\\_genome/GRCh38\\_full\\_analysis\\_set\\_plus\\_decoy\\_hla.fa](ftp://ftp.1000genomes.ebi.ac.uk/vol1/ftp/technical/reference/GRCh38_reference_genome/GRCh38_full_analysis_set_plus_decoy_hla.fa)

## Command lines

1. Alignment at lane level

```
bwa mem
  -Y
  -K 100000000
  -t 16
  -R $rg_string
  $reference_fasta
  $fastq_file(1)
  $fastq_file(2) | samtools view -Shb -o $bam_aligned -
```
2. Fix mate information in the BAM

```
java $jvm_args -jar picard.jar
  FixMateInformation
  MAX_RECORDS_IN_RAM=2000000
```

```
VALIDATION_STRINGENCY=SILENT
ADD_MATE_CIGAR=True
ASSUME_SORTED=true
I=$bam_aligned
O=$bam_fixmate
```

3. Merging lane level bam files to Sample level bam files

```
java $jvm_args -jar picard.jar
MergeSamFiles
USE_THREADING=true
MAX_RECORDS_IN_RAM=2000000
VALIDATION_STRINGENCY=SILENT
SORT_ORDER=queryname
INPUT=$bam_fixmate_1
INPUT=$bam_fixmate_2
OUTPUT=$bam_merged
```

4. Mark duplicates and coordinate sort BAM

```
java $jvm_args -jar picard.jar
MarkDuplicates
MAX_RECORDS_IN_RAM=2000000
VALIDATION_STRINGENCY=SILENT
M=$dedup_metrics
I=$bam_merged
O=$bam_dedup
```

```
java $jvm_args -jar picard.jar
SortSam
MAX_RECORDS_IN_RAM=2000000
VALIDATION_STRINGENCY=SILENT
SORT_ORDER=coordinate
CREATE_INDEX=true
I=$bam_dedup
O=$bam_sorted
```

5. Recalibrate base quality scores using known SNPs

```
java $jvm_args -jar GenomeAnalysisTK.jar
-T BaseRecalibrator
-downsample_to_fraction 0.1
-nct 4
--preserve_qscores_less_than 6
-L $autosomes
-R $reference_fasta
-o $recal_data.table
-I $bam_sorted
-knownSites $known_snps_from_dbSNP138
-knownSites $known_indels
-knownSites $known_indels_from_mills_1000genomes
```

```

java $jvm_args -jar GenomeAnalysisTK.jar
  -T PrintReads
  -nct 4
  --disable_indel_qual
  --preserve_qscores_less_than 6
  -SQQ 10
  -SQQ 20
  -SQQ 30
  -rf BadCigar
  -R $reference_fasta
  -o $bam_recalibrated
  -I $bam_sorted
  -BQSR $recal_data.table

```

## 6. Creating CRAM files

```

samtools view
  -C
  -T $reference_fasta
  -o $cram
  $bam_recalibrated

samtools index $cram

```

## 7. Raw variant calls using HaplotypeCaller on single sample

For variant discovery we used HaplotypeCaller in GVCF mode (Poplin et al., 2017) with sex-dependent ploidy settings on chromosome X and Y. Specifically, variant discovery on chrX was performed using diploid settings in females, diploid settings on PAR regions in males, and haploid settings on non-PAR regions in males. Variant discovery on chrY was performed with haploid settings in males and skipped in females.

```

java $jvm_args -jar GenomeAnalysisTK.jar
  -T HaplotypeCaller
  --genotyping_mode DISCOVERY
  -A AlleleBalanceBySample
  -A DepthPerAlleleBySample
  -A DepthPerSampleHC
  -A InbreedingCoeff
  -A MappingQualityZeroBySample
  -A StrandBiasBySample
  -A Coverage
  -A FisherStrand
  -A HaplotypeScore
  -A MappingQualityRankSumTest
  -A MappingQualityZero
  -A QualByDepth
  -A RMSMappingQuality
  -A ReadPosRankSumTest
  -A VariantType
  -l INFO

```

```

--emitRefConfidence GVCF
-rf BadCigar
--variant_index_parameter 128000
--variant_index_type LINEAR
-R $reference_fasta
-nct 1
-L $interval
-ploidy $ploidy
-I $bam_recalibrated
-o $gvcf

```

#### 8. Jointly recalibrate Genotype Quality score of all samples

```

java $jvm_args -jar GenomeAnalysisTK.jar
-T GenotypeGVCFs
-R $reference_fasta
-nt 5
--disable_auto_index_creation_and_locking_when_reading_rods
--variant $gvcf
-o $vcf_genotyped

```

#### 9. Variant Quality Score Recalibration (VQSR) to assign FILTER status

```

java $jvm_args -jar GenomeAnalysisTK.jar
-T VariantRecalibrator
-R $reference_fasta
-nt 5
-input $vcf_genotyped
-mode SNP
-recalFile $vqsr_snp.recal
-tranchesFile $vqsr_snp.tranches
-rscriptFile $vqsr_snp_plots.R
-resource:hapmap,known=false,training=true,truth=true,prior=15.0
$hapmap
-resource:omni,known=false,training=true,truth=true,prior=12.0
$kg_omni
-resource:1000G,known=false,training=true,truth=false,prior=10.0
$kg_snps
-resource:dbsnp,known=true,training=false,truth=false,prior=2.0
$dbsnps
-an QD
-an MQ
-an FS
-an MQRankSum
-an ReadPosRankSum
-an SOR
-an DP
-tranche 100.0
-tranche 99.8
-tranche 99.6
-tranche 99.4
-tranche 99.2

```

- tranche 99.0
- tranche 95.0
- tranche 90.0

```
java $jvm_args -jar GenomeAnalysisTK.jar
-T VariantRecalibrator
-R $reference_fasta
-nt 5
-input $vcf_genotyped
-mode INDEL
-recalFile $recalibrate_indel.recal
-tranchesFile $recalibrate_indel.tranches
-rscriptFile $recalibrate_indel_plots.R
-resource:mills,known=true,training=true,truth=true,prior=12.0
$kg_mills
-resource:dbsnp,known=true,training=false,truth=false,prior=2.0
$dbsnp
-an QD
-an FS
-an ReadPosRankSum
-an MQRankSum
-an SOR
-an DP
-tranche 100.0
-tranche 99.0
-tranche 95.0
-tranche 92.0
-tranche 90.0
--maxGaussians 4
```

```
java $jvm_args -jar GenomeAnalysisTK.jar
-T ApplyRecalibration
-R $reference_fasta
-nt 5
-input $vcf_genotyped
-mode SNP
--ts_filter_level 99.80
-recalFile $recalibrate_SNP.recal
-tranchesFile $recalibrate_SNP.tranches
-o $vcf_recalibrated_snp
```

```
java $jvm_args -jar GenomeAnalysisTK.jar
-T ApplyRecalibration
-R $reference_fasta
-nt 5
-input $vcf_recalibrated_snp
-mode INDEL
--ts_filter_level 99.0
-recalFile $recalibrate_INDEL.recal
```

```
-tranchesFile $recalibrate_INDEL.tranches
-o $vcf_recalibrated_snp_indel
```

## Definitions of delivered files

1. recalibrated\_variants.vcf.gz[.tbi]
  - All variants in Variant Call Format (VCF) file along with index.
2. recalibrated\_variants.annotated.vcf.gz[.tbi]
  - Normalized VCF stripped of genotype calls and annotated using [snpEff](#) and [BCFTools](#).
3. recalibrated\_variants.annotated.txt
  - Variant annotations in a tab-delimited file.
4. recalibrated\_variants.annotated.coding.txt
  - All annotated variants with HIGH/MODERATE impact in a tab-delimited file.
  - High impact - The variant is assumed to have high (disruptive) impact in the protein, probably causing protein truncation, loss of function or triggering nonsense mediated decay. e.g. stop\_gained, frameshift\_variant.
  - Moderate impact - A non-disruptive variant that might change protein effectiveness. e.g. missense\_variant, inframe\_deletion
5. recalibrated\_variants.annotated.coding\_rare.txt
  - All HIGH/MODERATE annotated variants with less than 5% allele frequency in 1000genomes and ExAC in a tab-delimited file.
6. recalibrated\_variants.annotated.clinical.txt
  - All low frequency HIGH/MODERATE annotated variants with possible clinical impact from [ClinVar](#) in a tab-delimited file.

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NA12376, NA12383, NA12386, NA12399, NA12400, NA12413, NA12414, NA12485, NA12489, NA12546, NA12707, NA12708, NA12716, NA12717, NA12718, NA12739, NA12740, NA12748, NA12749, NA12750, NA12751, NA12752, NA12753, NA12760, NA12761, NA12762, NA12763, NA12766, NA12767, NA12775, NA12776, NA12777, NA12778, NA12801, NA12802, NA12812, NA12813, NA12814, NA12815, NA12817, NA12818, NA12827, NA12828, NA12829, NA12830, NA12832, NA12842, NA12843, NA12864, NA12865, NA12872, NA12873, NA12874, NA12875, NA12877, NA12878, NA12889, NA12890, NA12891, NA12892]. The data were generated at the New York Genome Center with funds provided by NHGRI Grants 3UM1HG008901-03S1 and 3UM1HG008901-04S2.

If you have any questions please email [service@nygenome.org](mailto:service@nygenome.org).